MACHINE LEARNING FOR MIX DESIGN OF CONCRETE

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Objectives

Review the standard codes on mix design of concrete

Gather datasets of concrete composition for various grades of concrete

Choosing proper ML algorithms for the optimization and prediction of the ratio of materials for different grades of concrete

Create a User Interface(UI) and make these ratios accessible to everyone

Scope

The scope of ML in mix design of concrete is huge and it covers various stages from planning and material selection to the final performance evaluation. But the key areas are

- Prediction model for concrete properties
- Optimization of concrete mix proportions
- Data Analysis and knowledge extraction
- Generalization of this to all types of conditions
- Enhancing durability and estimation of the service life

Review on the literature

Civil engineering had already begun incorporating machine learning and the concrete industry is following the same path

We have been using ML algorithms in predicting the compressive strength of the concrete

This helps in reducing the wastage of materials and human power, i.e., reducing the cost of estimation

It also reduces the carbon footprint and reduces environment impact and move towards sustainability

Research Gaps

The availability of the data and its quality is a major research gap

Improving the accuracy of the ML algorithms selected for this purpose

Reducing the deviation of the compressive strength on site is another major challenge

Large scale validation of these results is to be done before relying on this completely

Works Done

Completed the study of standard codes for concrete mix design, including IS 10262:2009 and IS 10262:2019, along with relevant American and British standard codes

Collected some of the datasets of material composition of concrete

Constraints

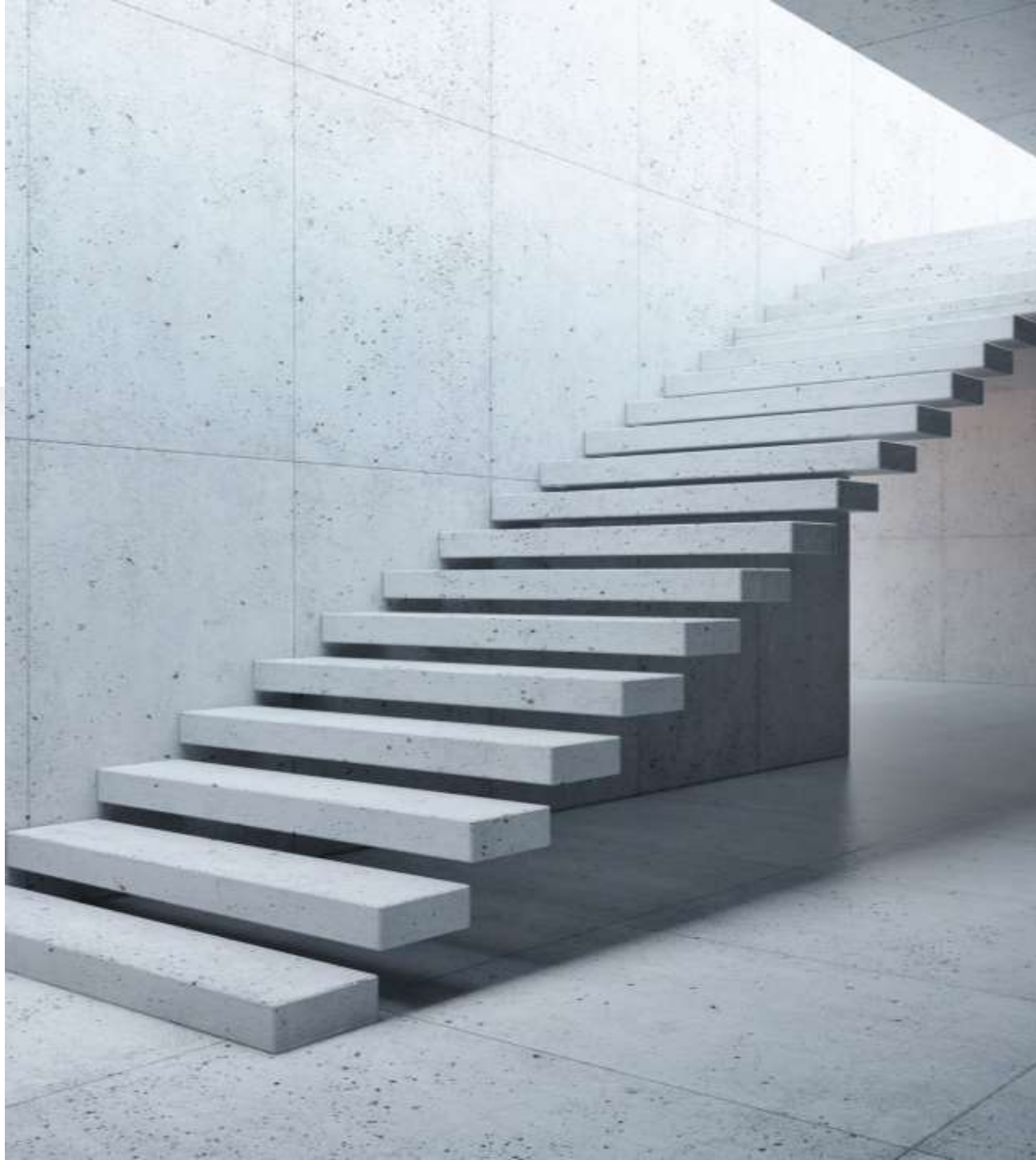
Availability of the data of material composition of concrete on the internet is very much limited

Feature selection is another major constraint faced in this project as this involves many parameters such as cement type, w/c ratio, aggregates, etc., have equal importance in determining the compressive strength of the concrete

Selecting the proper ML model is important to avoid the risk of overfitting

References

- IS 10262:2009 and IS 10262:2019
- Machine Learning Techniques in Concrete Mix Design Patryk Ziolkowski and Maciej Niedostatkiewicz
- Concrete Mix Design by IS, ACI and BS Methods: A Comparative Analysis Ashish Chhachhia



Thank You